REQUIREMENTS TRACE MATRIX

Document Ref: Appendix 01_Functional Requirements_LIMS - Thermo Fisher
Version: 1
Date: 04 June 2024

gathering status	Module	Req. No.	URS Ref.	Delivery Phase	Functional Area / Workshop	Requirement	Priority	Type	Estimate
complete			4.1.1.1		Sample Login/Sample Receipt	The system shall provide functionality to log samples using configurable templates In case of plants with Smart factory, Smart factory MES samples are collected in those application for standard testing Samples have to be created using QCLaD for non-standard testing, and for sites where there is no MES or Smart factory.	Must		
complete	Sample	R001	4.1.1.1.1	Phase A		System shall be able to login samples using configurable templates	Must	Configuration	
complete			4.1.1.1.2		Mark	Henkel offer the sample request template for reference.			
complete	Sample	R002	4.1.1.1.3	Phase A		System shall be able to create samples manually for standard testing.	Must	Configuration	
complete	Sample	R003	4.1.1.1.4	Phase A		System shall be able to create samples manually for non-standard testing.	Must	Configuration	
complete			4.1.1.2			The system shall provide functionality to log samples automatically based on trigger events in an external system or schedule	Must		
complete	Interface	R004	4.1.1.2.1	Phase A		System shall be able to receive inspection lot from SAP and create sample and test. Refer to 4.5.1.1		Customisation	
complete	Interface	R005	4.1.1.2.2	Phase B		System shall be able to trigger a resample from MES.		Customisation	
complete	Sample	R006	4.1.1.2.2	Phase A		System shall be able to create sample based on events scheduled within QCLAD.		Standard	
complete			4.1.1.3			The system shall assign a unique sample ID to each sample logged in Unique id has to be pulled from sample label if exists and reuse. If it doesn’ t exist, system can create one.	Must		
complete	Sample	R007	4.1.1.3.1	Phase A		System shall assign a unique sample ID to each sample logged in.	Must	Configuration	
complete	Sample	R008	4.1.1.3.2	Phase A		System shall be able to scan the Labels from MES system and fetch unique ID.		Customisation	
complete			4.1.1.3.3		mark	Henkel need to check label template with Dragon. (offer QR code rule)			
complete	Sample	R009	4.1.1.3.4	Phase A		System shall allow user to print QCLaD Labels if the Plant doesn’ t use MES system.	Must	Standard	
complete			4.1.1.3.5		mark	Henkel already offered sample ID naming rules for non-standard Sample .(TF already provided sample ID naming rules examples)			
complete			4.1.1.4			The system shall provide functionality to capture the date and time of sample receipt	Must		
complete	Sample	R010	4.1.1.4.1	Phase A		System shall be able to retrive the sample receive date and time.	Must	Standard	
complete			4.1.1.5			The system shall provide functionality to allow privileged users to cancel samples from the system	Must		
complete	Sample	R011	4.1.1.5.1	Phase A		System shall be able to allow privileged users to cancel samples.	Must	Standard	
complete	Sample	R012	4.1.1.5.2	Phase A		System shall be able to add a mandatory comment for the cancellation.		Configuration	

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Document Ref: Appendix 01_Functional Requirements_LIMS - Thermo Fisher
Version: 1
Date: 04 June 2024

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complete			4.1.1.6			The system shall provide functionality to define configurable attributes for a sample which can be modified	Must		
complete	Sample	R013	4.1.1.6.1	Phase A		System shall be able to define configurable attributes for a sample which can be modified.	Must	Standard	
complete			4.1.1.7			The system shall provide functionality to allow the definition of sample types and sampling plans. This should include functionality to log in predefined batches of samples at one time in accordance with a defined sampling plan	Must		
complete	Sample	R014	4.1.1.7.1	Phase A		System shall be able to allow the definition of sample types and test schedule.	Must	Standard	
complete			4.1.1.7.2		Mark	Henkel need to check function of sample typle and sampling plans with Maline. Re: sampling plan means sampling quantity			
complete	Sample	R015	4.1.1.7.3	Phase A		System shall be able to log in predefined batches of samples at one time in accordance with a defined sampling plan.	Must	Configuration	
complete			4.1.1.8			The system shall provide functionality to log in either single or multiple	Must		
complete	Sample	R016	4.1.1.8.1	Phase A		System shall be able to log in either single or multiple samples from the same batch.	Must	Standard	
complete			4.1.1.9			The system shall provide functionality to manage ‘sample split’	Must		
complete	Sample	R017	4.1.1.9.1	Phase A		The system shall be able to split samples.	Must	Standard	
complete	Security	R018	4.1.1.9.2	Phase A		System shall be able to display the traceability of split samples with a parent sample.		Standard	
complete			4.1.1.10			The system shall provide functionality to log non routine samples with the user selecting a pre-defined specification or manually assigning tests from a pre- defined test list	Must		
complete	Sample	R019	4.1.1.10.1	Phase A		System shall be able to log in non routine samples.	Must	Standard	
complete	Sample	R020	4.1.1.10.2	Phase A		System shall allow specific user to select specifications and test methods for non routine samples.	Must	Standard	
complete			4.1.1.11			The system shall provide functionality to manage retesting of samples that have an adverse finding	Must		
complete	Sample	R021	4.1.1.11.1	Phase A		system shall be able to retest.	Must	Configuration	
complete	Sample	R022	4.1.1.11.2	Phase A		System shall be record all failure as well as success of parameters in QCLAD.	Must	Standard	
complete	Interface	R023	4.1.1.11.3	Phase A		System shall be push all failure as well as success of parameters to SAP. Refer to 4.5.1.7		Customisation	
complete			4.1.1.12			The system shall provide functionality to manage re-samples and their association with the original sample	Must		
complete	Sample	R024	4.1.1.12.1	Phase A		System shall be able to resample and association with the original sample.	Must	Standard	
complete			4.1.1.13			The system shall provide functionality to associate specification limits to the tests for a sample	Must		

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Document Ref: Appendix 01_Functional Requirements_LIMS - Thermo Fisher
Version: 1
Date: 04 June 2024

gathering status	Module	Req. No.	URS Ref.	Delivery Phase	Functional Area / Workshop	Requirement	Priority	Type	Estimate
complete	Sample	R025	4.1.1.13.1	Phase A		System shall be able to associate MLP to the tests for a sample.	Must	Standard	
complete	Interface	R026	4.1.1.13.2	Phase A		System shall be receive specification limit to be pulled from SAP. Refer to 4.5.1.3		Customisation	
complete			4.1.1.14			The system shall provide functionality to allow restriction of sample types or sample groups to specific user groups, specialist analysts and restriction can be modified during the sample life cycle	Must		
complete	Security	R027	4.1.1.14.1	Phase A		System shall be able to control data access by group.	Must	Standard	
complete	Sample	R028	4.1.1.14.2	Phase A		System shall support modify specialist analysts and group restriction during the sample life cycle.	Must	Configuration	
complete			4.1.1.15			The system shall provide the capability to enter free text information at Sample log-in	Must		
complete	Sample	R029	4.1.1.15.1	Phase A		System shall be able to allow user to enter comments at Sample log-in.	Must	Configuration	
complete			4.1.1.16			The system must be able to scan barcode information and transmit this information to the solution using an appropriate system	Must		
complete	Sample	R030	4.1.1.16.1	Phase A		System shall be able to scan barcode or QRCode and match barcode or QRCode sample in QCLAD system.		Customisation	
complete			4.1.1.17			The system should be able to automatically login samples from a schedule or other systems (SAP, Henkel Smart Factory, MES)	Must		
complete	Sample	R031	4.1.1.17.1	Phase A		System shall be automatically login samples from a schedule.	Must	Standard	
complete	Interface	R032	4.1.1.17.2	Phase A		System shall be automatically login samples from SAP. Refer to 4.5.1.1		Customisation	
complete	Interface	R033	4.1.1.17.3	Phase B		System shall be automatically login samples from MES.		Customisation	
complete			4.1.1.18			The system shall provide functionality to allow entry / modification of attribute data at point of single or multiple sample receipt using configurable templates. For sites where there is labeling through MES, SFS or labelling solution, sample receipt has to be done by scanning the labels. For non MES/SFS sites, labels created from QCLAD to be used for scanning the sample receipt	Must		
complete	Sample	R034	4.1.1.18.1	Phase A		Sytem shall be able to allow entry / modification of attribute data at point of single or multiple sample receipt using configurable templates.	Must	Configuration	
complete	Sample	R035	4.1.1.18.2	Phase A		System shall be able to receive sample by scanning MES label		Customisation	
complete	Report&Label	R036	4.1.1.18.3	Phase A		System shall be able to create labels for sample scan receiving for non MES samples.	Must	Configuration	
complete			4.1.1.19			The system shall provide functionality to allow entry of any irregularities recorded and route sample according to laboratory business rules	Must		

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Document Ref: Appendix 01_Functional Requirements_LIMS - Thermo Fisher
Version: 1
Date: 04 June 2024

gathering status	Module	Req. No.	URS Ref.	Delivery Phase	Functional Area / Workshop	Requirement	Priority	Type	Estimate
pending	Sample	R037	4.1.1.19.1	Phase A	Mark	The system shall provide functionality to allow entry of any irregularities recorded sample according to laboratory business rules.		Standard	
pending	Sample	R038	4.1.1.19.2	Phase A		The system shall provide functionality to allow entry of any routine sample according to laboratory business rules.			
complete			4.1.1.20			The system shall provide functionality to allow users to change sample attributes after sample receipt	Must		
complete	Sample		4.1.1.20.1	Phase A		System shall allow users to modify sample attributes after sample receipt.	Must	Configuration	
complete			4.1.2.1		Sample Testing/Results Entry	The system shall provide two ways to assign works/tests to users: i) Manually, Lab supervisor can allocate the work periodically based on requirements. ii) Automatically, QCLaD arrange work based on pre-defined rules. The rule is Henkel customize, it can be configurable by Henkel site QC supervisor. The rule is based on Henkel resource (QC technician skill, workload, work shift and instrument availability, etc.); iii) function to support QC supervisor stop or modify auto assign task. iv) capability to receive QC technician skills from Henkel Skill Matrix application for automatic task assignment in case the plant has configured Skill matrix application	Must		
complete	Assignment		4.1.2.1.1	Phase A		System shall be able to assign tests to an operator manually or automatically.		Customisation	
complete	Assignment		4.1.2.1.2	Phase A		Automatically assigning details refer to 4.2.1.1 part.		Customisation	
complete	Assignment		4.1.2.1.3	Phase A		System shall support to stop or modify auto assign task.		Customisation	
complete	Assignment		4.1.2.1.4	Phase B		System shall be able to assign test based on the versatility matrix from OSDP system once integrated.		Customisation	
complete			4.1.2.3			The system shall provide functionality to reference all results data to the sample	Must		
complete	Sample		4.1.2.3.1	Phase A		System shall be able to display all results data to the sample.	Must	Standard	
complete	Interface		4.1.2.3.2	Phase A		System shall be able to retrive the result data in CoA data already available in SAP inspection lot.		Customisation	
complete	Interface		4.1.2.3.3	Phase A		System shall be able to display the Data recorded within inspection lot, MIC, Spec, Test result from the CoA in SAP even if the data entry is not required in QCLAD.		Customisation	
complete			4.1.2.4			The system shall provide functionality to calculate results based on reported or recorded data entered against a sample and data obtained from	Must		

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Document Ref: Appendix 01_Functional Requirements_LIMS - Thermo Fisher
Version: 1
Date: 04 June 2024

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complete	Sample		4.1.2.4.1	Phase A		System shall be able to calculate results based on reported or recorded data which from instruments.	Must	Configuration	
complete			4.1.2.5			The system shall provide functionality to automatically check intermediate values (e.g., raw, recorded and reported data) and final results against specification limits and flag failures Intermediate values: all data in whole testing process, e.g., raw data captured from instrument, recorded-data, data calculated by formular, reported data, data which push to SAP	Must		
complete	Sample		4.1.2.5.1	Phase A		System shall be able to record all data in whole testing process, including raw data captured from instrument, entry data, data calculated by formular, reported data, data which push to SAP.		Standard	
complete	Sample		4.1.2.5.2	Phase A		System shall be able to mark final results against specification limits and flag failures.	Must	Configuration	
complete			4.1.2.6			The system shall provide functionality to manage repeat testing	Must		
complete	Interface		4.1.2.6.1	Phase A		System shall be able to support to manage repeat testing.(e.g. entry multiple results and then get max or min or average value)		Customisation	
complete			4.1.2.7			The system shall provide functionality to enable results entry for multiple tests performed on a single sample	Must		
complete	Sample		4.1.2.7.1	Phase A		System shall be able to support results entry for multiple tests performed on a single sample.	Must	Standard	
complete			4.1.2.8			The system shall provide functionality to enable results entry for an individual test on a single sample	Must		
complete	Sample		4.1.2.8.1	Phase A		System shall be able to support results entry for an individual test on a	Must	Standard	
complete			4.1.2.9			The system shall provide functionality to enable results entry for single or multiple tests performed on multiple samples	Must		
complete	Sample		4.1.29.1	Phase A		System shall be able to support results entry for single or multiple tests performed on multiple samples.	Must	Standard	
complete			4.1.2.10			The system shall provide functionality to verify entered results data (e.g., pH, sample color, specific gravity) against pre-set method limits E.g., PH pre-set method limits is 0-14, result cannot be 15	Must		
complete	Sample		4.1.2.10.1	Phase A		System shall be able to verify entered results data (e.g., pH, sample color, specific gravity) against specification limits.	Must	Standard	
complete			4.1.2.11			The system shall prevent the acceptance of partial or incomplete data entry for results and samples E.g., one test has two parallel samples, system doesn’ t allow to report data for only for one sample.	Must		
complete	Sample		4.1.2.11.1	Phase A		System shall be able to prevent the acceptance of partial or incomplete data entry for results and active samples.	Must	Configuration	
complete			4.1.2.12			The system shall provide configurable screens aligned to analytical method steps with potential to lead the user through the operation in a step-by-step manner. This would facilitate data capture in real time Can record time taken as well and have video instruction about how to do test.	Must		

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Document Ref: Appendix 01_Functional Requirements_LIMS - Thermo Fisher
Version: 1
Date: 04 June 2024

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complete	LES		4.1.2.12.1	Phase B		System shall be able to provide configurable analytical method steps with potential to lead the user through the operation in a step-by-step manner.(LES module)	Must	Standard	
complete	IM Integartion		4.1.2.12.2	Phase B		System shall be able to capture instrument data(for Type A) and report from instrument (for Type B) in real time.	Must	Standard	
complete	LES		4.1.2.12.3	Phase B		System shall be able to upload attachements(including video) in LES.	Must	Standard	
complete	LES		4.1.2.12.4	Phase B		System shall be able to record start date and end date in LES.	Must	Standard	
			4.1.2.12.5		Mark	Video Size depends on Henkel service provided performance.			
complete			4.1.2.13			The system shall be able to accept numeric, qualified numeric, free text, characters, calculated values, file, Boolean and pre-defined list data entry	Must		
complete	Sample		4.1.2.13.1	Phase A		System shall be able to support multipe analysis type, including numeric, text, characters, calculated, file, Boolean and optional.	Must	Standard	
complete			4.1.2.14			The system shall have the capability to receive data via manual input (for type C and D instrument and when type A and B malfunction), electronic transfer and instrument interface and indicate mode of entry Instrument interface: get data from instrument directly.	Must		
complete	Sample		4.1.2.14.1	Phase A		System shall be able to entry result data manually.	Must	Standard	
	IM Integartion		4.1.2.14.2	Phase A		System shall be able to entry result data via instrument automatically.		Configuration	
complete			4.1.2.15			The system shall provide a way to manually upload files (Excel, CSV, Xps, xml, Jpg or Pdf format) for documentation	Must		
complete	Sample		4.1.2.15.1	Phase A		System shall support attachment format: Xps, txt, resx, repx, xml, rtf, jpg, jpeg, img, svg, doc, pdf, ico, png, lbl, resources, caf, csv, log, msg,	Must	Standard	
complete			4.1.2.16			The system shall be able to interface with the Henkel type A and B instruments.	Must		
complete	IM Integartion		4.1.2.16.1	Phase A		System shall be able to interface with the Henkel type A and B instruments.		Configuration	
complete			4.1.2.17			The system shall be able to automatically capture results from the Henkel type A and B instruments	Must		
complete	IM Integartion		4.1.2.17.1	Phase A		System shall be able to automatically capture results from the Henkel type A and B instruments.		Configuration	
complete			4.1.2.18			The system should have capability to map the equipment based on test. The equipment should be selected by system based on the test and equipment master data In case of any change, the user can modify based on access	Must		
complete	LES		4.1.2.18.1	Phase B		System shall be able to selecte equipments based on test method in LES.	Must	Standard	

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Document Ref: Appendix 01_Functional Requirements_LIMS - Thermo Fisher
Version: 1
Date: 04 June 2024

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complete	LES		4.1.2.18.2	Phase B		System shall be able to allow authorized user to modify instruments in LES.	Must	Standard	
complete			4.1.2.19			The system shall be able to protect the calculation template(s) for the final test results so that they cannot be changed by all users The calculation template(s) to be controlled by user role/rights access	Must		
complete	LES		4.1.2.19.1	Phase B		System shall only allow authorized users to modify calculation template in LES.	Must	Standard	
complete	Security		4.1.2.19.2	Phase B		System shall support audit traceability in LES.		Standard	
complete			4.1.2.20			The system shall record the source of any entered data (e.g., instrument ID) and / or the user ID of any user entering data	Must		
complete	LES		4.1.2.20.1	Phase B		The system shall record the source of any entered data (e.g., instrument ID) and / or the user ID of any user entering data.	Must	Configuration	
complete			4.1.2.21			The system shall provide the functionality to record multiple users against a test or result (e.g., if a test is performed by two analysts) E.g., OHV test, GAP value and Acid value needed for OHV test, QCLaD allows different person conduct GAP and Acid value at same time and record data in QCLaD.	Must		
complete	Sample		4.1.2.21.1	Phase A		System shall be able to record multiple users against a test or result.	Must	Standard	
complete			4.1.2.22			The system shall manage retesting without loss of original data	Must		
complete	Sample		4.1.2.22.1	Phase A		System shall be able to retest without loss of original data.	Must	Configuration	
complete			4.1.2.23			The system shall enforce entry of a justification if there is a retest / repeat	Must		
complete	Sample		4.1.2.23.1	Phase A		System shall be able to enforce entry of a justification if there is a retest/resample.	Must	Configuration	
complete	Sample		4.1.2.23.2	Phase A		system shall be able to record comment as mandatory and it is configurable.	Must	Configuration	
complete			4.1.2.24			The system shall provide functionality to perform simple trend analysis at Results Entry to identify any atypical results and alert the user accordingly	Wont	Out Of Scope	
complete			4.1.2.24.1		Mark	SQC is not in scope. Henkel need to check with Anson.	Wont	Out Of Scope	
complete			4.1.2.25			The system shall be configurable, at the test level, to enforce peer review by a different user to the user that entered the test results	Must		
complete	Sample		4.1.2.25.1	Phase A		System shall be able to enforce peer review by a different user at the material group level.	Must	Configuration	
complete	Sample		4.1.2.25.2	Phase A		System shall be able to support peer review by configurable option.	Must	Configuration	
complete			4.1.2.26			The system shall allow the peer review of results by Single sample, Multi test and Multi sample, single test where required	Must		

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Document Ref: Appendix 01_Functional Requirements_LIMS - Thermo Fisher
Version: 1
Date: 04 June 2024

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complete	Sample		4.1.2.26.1	Phase A		System shall allow the peer review of results by Single sample, Multi test and Multi sample, single test where required.	Must	Standard	
complete			4.1.2.27			The system must lock test data after peer review to prevent further modification except with appropriate permissions and audit trail	Must		
complete	Sample		4.1.2.27.1	Phase A		System shall prevent user from further modification on test data after peer review except authorized user and audit trail will be record.	Must	Configuration	
complete			4.1.2.28			The system shall provide planning tools to display and assess laboratory workload, including analyst, sample, instrument assignment and sample testing schedule.	Must		
pending			4.1.2.28.1		Mark	Henkel need to offer dashboard template.			
pending	Dashboard		4.1.2.28.2			System shall provide dashboard.		Customisation	
complete			4.1.3.1		Authorization	The system shall provide configurable functionality to approve individual	Must		
complete	Sample		4.1.3.1.1	Phase A		System shall be able to approve individual results.	Must	Standard	
complete	Sample		4.1.3.1.2	Phase A		System shall be able to support configure approve option within workflow.	Must	Configuration	
complete			4.1.3.2			The system shall allow for the approval of single or multiple samples	Must		
complete	Sample		4.1.3.2.1	Phase A		System shall be able to approve single or multiple samples.	Must	Standard	
complete			4.1.3.3			The system shall prevent users that have peer reviewed results from changing any data on the sample when authorize	Must		
complete	Sample		4.1.3.3.1	Phase A		System shall prevent user from further modification on test data after peer	Must	Configuration	
complete	Sample		4.1.3.3.2	Phase A		System shall be able to return test to tester if result need modification .	Must	Configuration	
complete	Sample		4.1.3.3.3	Phase A		System shall be able to support return test to another operator.		Customisation	
complete			4.1.3.4			The system shall provide configurable system approval functionality to automatically assign an authorization at test level and sample level.	Must		
complete	Sample		4.1.3.4.1	Phase A		System shall be able to configure system approval functionality at test level and sample level.	Must	Configuration	
complete			4.1.3.5			The system shall provide functionality for manual approval at test level,	Must		
complete	Sample		4.1.3.5.1	Phase A		System shall be able to approval manually at test level and sample level.	Must	Configuration	
complete			4.1.3.6			The system shall prevent the sample authorization with incomplete testing and alert defined users to the situation Prevent the sample authorization means if the sample in one incomplete phase (e.g., Testing), it cannot go to next phase. Sample can only in one phase in QCLaD.	Must		
complete	Sample		4.1.3.6.1	Phase B		System shall prevent the sample authorization with incomplete testing and alert defined users to the situation in folder.	Must	Configuration	
complete	Sample		4.1.3.6.2	Phase A		System shall only have one status of a sample in one time.	Must	Standard	
complete			4.1.3.7			The system shall allow authorized users to change the status of a sample to allow controlled modifications or entry of data E.g., QCLaD allows user to modify inspection plan of non-standard testing, inspection plan of standard test should come from SAP.	Must		

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Document Ref: Appendix 01_Functional Requirements_LIMS - Thermo Fisher
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Date: 04 June 2024

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complete	Sample		4.1.3.7.1	Phase A		System shall allow authorized users to change the status of a sample to allow controlled modifications or entry data.	Must	Configuration	
complete			4.1.3.8			The system shall allow certain authorized users to change the proposed user for testing	Must		
complete	Sample		4.1.3.8.1	Phase A		System shall allow certain authorized users to change the proposed user for	Must	Configuration	
complete			4.1.3.9			The system shall provide a way to check and indicate the results/samples waiting for approval.	Must		
complete	Sample		4.1.3.9.1	Phase A		System shall provide a way to check and indicate the results/samples waiting for approval.	Must	Configuration	
complete			4.1.4.1		Reporting	The system shall provide a reporting tool to have users to define report templates by themselves	Must		
complete	Report&Label		4.1.4.1.1	Phase A		System shall allow users to define report templates by themselves using reporting tool.	Must	Standard	
complete			4.1.4.10			The system shall support to generate single sample report, multiple sample report, COA report etc.	Must		
complete	Report&Label		4.1.4.10.1	Phase A		System shall support to generate single sample report, multiple sample report, COA report.	Must	Configuration	
complete			4.1.4.10.2		Mark	Henkel need to offer all report template.			
complete			4.1.4.11			The system shall provide functionality to route the output of reports to screen, printer, file (CSV, XML) or email	Must		
	Report&Label		4.1.4.11.1	Phase A		System shall be able to route the output of reports to screen, printer, file (CSV, XML, PDF) or email.	Must	Standard	
complete			4.1.4.12			The system shall provide functionality to generate reports in various formats (e.g., tabular, matrix, graphical or text)	Must		
complete	Report&Label		4.1.4.12.1	Phase A		System shall be able to support various formats (e.g., tabular, matrix, graphical or text) in report templates.	Must	Standard	
complete			4.1.4.13			The systems performance must not be adversely affected by running reports	Must		
complete	Report&Label		4.1.4.13.1	Phase A		Systems performance must not be adversely affected by running reports.	Must	Standard	
complete	Report&Label		4.1.4.13.2	Phase A		System shall be able to give warning of response if user selects a range which will take time.		Customisation	
complete			4.1.4.14			The system shall be capable of sending notifications and alert messages via electronic mail	Must		
complete	Sample		4.1.4.14.1	Phase B		System shall be able to send notifications and alert messages within folders.	Must	Configuration	
complete			4.1.4.15			The system shall be capable to generate Henkel customer requested CoA (certificate of analysis) report with customer' s template based on data in QCLaD system. The system should have capability to allow site admin to create customer CoA template based on different customer requirements. The system should have capability to send email with attachment to dedicate E-mail address based on master data configuration	Must		
complete	Report&Label		4.1.4.15.1	Phase A		System shall allow users to define report templates based on customer or product .	Must	Standard	
complete	Report&Label		4.1.4.15.2	Phase A		System shall allow user to create and modify report templates.	Must	Standard	
complete	Report&Label		4.1.4.15.3	Phase A		System shall allow user to update version for reprot templates.	Must	Standard	

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Document Ref: Appendix 01_Functional Requirements_LIMS - Thermo Fisher
Version: 1
Date: 04 June 2024

gathering status	Module	Req. No.	URS Ref.	Delivery Phase	Functional Area / Workshop	Requirement	Priority	Type	Estimate
complete	Notification		4.1.4.15.4	Phase B		System shall be able to send email with attachment to dedicate E-mail address based on master data configuration or store the file in a configurable shared drive automatically.		Customisation	
complete			4.1.4.2			The text style or format in the text report shall be edible by the users.	Must		
complete	Report&Label		4.1.4.2.1	Phase A		System shall be able to provide editable text style or format in the text report.	Must	Standard	
complete			4.1.4.3			The system must be able to report on the analyzed data at all levels (e.g., raw, recorded and reported analytical data)	Must		
complete	Report&Label		4.1.4.3.1	Phase A		System must be able to report on the analyzed data at all levels (e.g., raw, recorded and reported analytical data).	Must	Configuration	
complete			4.1.4.4			The system shall collate results from multiple tests for each sample	Must		
complete	Report&Label		4.1.4.4.1	Phase A		System shall collate results from multiple tests for each sample.	Must	Configuration	
complete			4.1.4.5			The system shall be able to support to insert electrical signature, company logo, picture, trending chart, table etc. in the reports.	Must		
complete	Report&Label		4.1.4.5.1	Phase A		System shall be able to support to insert signature picture, company logo, picture, trending chart, table in the reports.	Must	Standard	
complete			4.1.4.5.2			System shall be able to support end app insert handwritting signature by pad.	Wont	Out Of Scope	
complete			4.1.4.6			The system shall provide functionality to manually or automatically trigger	Must		
complete	Sample		4.1.4.6.1	Phase A		System shall provide functionality to manually or automatically trigger a report in response to workflow events.		Configuration	
complete			4.1.4.7			The system shall provide a library of pre-defined controlled reports	Must		
complete	Report&Label		4.1.4.7.1	Phase A		System shall be able to provide a library of report templates.	Must	Configuration	
complete			4.1.4.8			The system shall provide ability to generate analytical reports based on the data recorded in QCLAD like date time, QC operators details along with	Must		
complete	Report&Label		4.1.4.8.1	Phase A		System shall be able to generate analytical reports based on the data recorded in QCLAD like date time, QC operators details along with inspection.	Must	Configuration	
complete			4.1.4.9			The system shall provide a way to generate and save "Test Report" template for generation and printing test report of other samples	Must		
complete	Report&Label		4.1.4.9.1	Phase A		System shall be able to generate and save "Test Report" template.	Must	Configuration	
complete	Report&Label		4.1.4.9.2	Phase A		System shall be able to print test report of other samples based on templates.	Must	Standard	
complete			4.1.5.1		Henkel Sample Type	The system should be capable to manage all sample types in Henkel, including, Standard test sample to support production: raw material sample (ROH), semi-finish goods sample/ bulk sample (UNB & UNF), finish goods sample (FER). Non-standard test sample: consumable, package material, instrument	Must		
complete	Sample		4.1.5.1.1	Phase A		System shall be able to manage Standard RM test for Incoming workflow.		Configuration	
complete	Sample		4.1.5.1.2	Phase A		System shall be able to manage Standard in process testing for semi finished and finished goods workflow.		Configuration	

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Document Ref: Appendix 01_Functional Requirements_LIMS - Thermo Fisher
Version: 1
Date: 04 June 2024

gathering status	Module	Req. No.	URS Ref.	Delivery Phase	Functional Area / Workshop	Requirement	Priority	Type	Estimate
complete	Sample		4.1.5.1.3	Phase A		System shall be able to manage Standard finished goods outgoing test workflow.		Configuration	
complete	Sample		4.1.5.1.4	Phase A		System shall be able to manage Consumables incoming test workflow.		Configuration	
complete	Sample		4.1.5.1.5	Phase A		System shall be able to manage package material incoming test workflow.		Configuration	
complete	Sample		4.1.5.1.6	Phase A		System shall be able to manage instrument calibration sample test workflow.		Configuration	
complete	Sample		4.1.5.1.7	Phase A		System shall be able to manage test for MSA workflow.		Configuration	
complete	Sample		4.1.5.1.8	Phase A		System shall be able to manage Certification test for QC technician workflow.		Configuration	
complete	Sample		4.1.5.1.9	Phase A		System shall be able to manage Test for SCMS/NCMR /CCMS investigation workflow.		Configuration	
complete	Sample		4.1.5.1.10	Phase A		System shall be able to manage Test for NCMR-Shelf life extension (RM, SFG, FG, Package material, consumble & Instrument calibration sample) workflow.		Configuration	
complete	Sample		4.1.5.1.11	Phase A		System shall be able to manage Testing for Customer/Supplier/PD/TCS/Other		Configuration	
complete	Sample		4.1.5.1.12	Phase A		System shall be able to manage Moving sample/material to storage location		Configuration	
complete	Sample		4.1.5.1.13	Phase A		System shall be able to manage Perform skip lot testing workflow.		Configuration	
complete			4.1.6.1		Non-testing work requirements	The system should be capable to maintain/arrange storage location of QC retain sample, consumables (reagent, etc.), spare parts. The system should be capable to maintain safety stock of consumables and spare parts, etc. The system should have field to maintain storage condition for retain samples, consumables and spare parts. Send auto notification once the sample near expiration date based on configurable rule. Send auto notification once it' s below safety stock of consumables and spare parts based on configurable rule.	Must		
complete	Sample		4.1.6.1.1	Phase A		System shall be able to maintain location.	Must	Standard	
complete	Stock		4.1.6.1.2	Phase A		System shall be able to support QC retain sample, consumables and spare parts can select location.	Must	Configuration	
complete	Stock		4.1.6.1.3	Phase A		System shall be able to maintain safety stock of consumables and spare parts.	Must	Standard	
complete	Stock		4.1.6.1.4	Phase A		System shall be able to maintain storage condition for retain samples, consumables and spare parts.	Must	Configuration	
complete	Dashboard		4.1.6.1.5	Phase A		System shall support dashboard to display and mark in color once the sample near expiration date based on configurable rule.		Customisation	
complete	Dashboard		4.1.6.1.6	Phase A		System shall support dashboard to display and mark in color once it' s below safety stock of consumables and spare parts based on configurable rule.		Customisation	
complete			4.1.6.1.7		Mark	Henkel need to offer these dashboard templates.			

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Document Ref: Appendix 01_Functional Requirements_LIMS - Thermo Fisher
Version: 1
Date: 04 June 2024

gathering status	Module	Req. No.	URS Ref.	Delivery Phase	Functional Area / Workshop	Requirement	Priority	Type	Estimate
complete			4.1.7.1		Special testing requirement management	The system should have capability to manage special test request (Testing for Customer/Supplier/PD/AE/Other sites/MOC/NPI / Improvement project, etc.) The system shall be capable of allow QCLaD user and non-QCLaD users to raise test request. Non-QCLaD user should have required user credentials. The system shall have capability to allow user to select which plant to conduct the test, which test needed, when is the expected complete date, etc. System should have capability to allow QC supervisor to approve the test requirement. System should have capability to enter results, report results, generate test report and send test report to requestor' s E-mail address. System should have capability to track the test status, analysis test workload and show it on Dashboard.	Must		
complete	Sample		4.1.7.1.1	Phase A		System shall be able to manage external test request using non-standard testing workflow.(4.1.5.1.4-4.1.5.13)		Customisation	
complete	Sample		4.1.7.1.2	Phase A		System shall be able to allow QCLaD users to raise test request.		Configuration	
pending			4.1.7.1.3		Mark	System shall be able to allow non-QCLaD users to raise test request(Both side need further discussion with Neelima)			
complete	Sample		4.1.7.1.4	Phase A		System shall be able to use template to allow user to select plant, analysis,expected complete date and comments.		Configuration	
complete			4.1.7.1.5		Mark	Henkel need to offer testing request templates.			
complete	Sample		4.1.7.1.6	Phase A		System shall be able to allow QC supervisor to approve the test requirement.	Must	Configuration	
complete	Sample		4.1.7.1.7	Phase B		System shall be able to enter results, report results, generate test report and send test report to requestor' s E-mail address.		Customisation	
complete	Dashboard		4.1.7.1.8	Phase A		System shall be able to track the test status, analysis test workload and show it on Dashboard.		Customisation	
complete			4.1.7.1.9		Mark	Henkel want to check the details of the test status. System shall support this.			
complete			4.1.7.1.10		Mark	Henkel need to offer Dashboard template about test status, analysis test workload.			
complete			4.2.1.1		Schedular	Sample test schedule arrangement way: 1. QC supervisor manual assign task to a user; 2. QCLaD automatic assign task to a user based configured rule (pending test items of standard and non-standard sample, user testing skill, user and test instrument availability, sample status, SAP test skip lot rules, other pending QC works, e.g., calibration, MSA works, etc.).	Must		
complete	Sample		4.2.1.1.1	Phase A		System shall be able to have option for a sample preparation which may be required for some analysis.	Must	Standard	
complete	Sample		4.2.1.1.2	Phase A		System shall be able to assign preparation to an operator.	Must	Customisation	
complete	Sample		4.2.1.1.3	Phase A		System shall be able to have multiple preparaiton for a analysis.		Customisation	
complete	Sample		4.2.1.1.4	Phase A		System shall be able to assign multiple instruments for preparation.		Customisation	
complete	Sample		4.2.1.1.5	Phase A		System shall be able to assign test to an operator for each test.	Must	Customisation	

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Document Ref: Appendix 01_Functional Requirements_LIMS - Thermo Fisher
Version: 1
Date: 04 June 2024

gathering status	Module	Req. No.	URS Ref.	Delivery Phase	Functional Area / Workshop	Requirement	Priority	Type	Estimate
complete	Assignment		4.2.1.1.6	Phase A		System shall be able to have standard time defined for each test assignment. This shall come from the master data to LIMS.	Must	Configuration	
complete	Sample		4.2.1.1.7	Phase A		System shall allow to assign instrument from the list available for the lab for each test by the operator.	Must	Configuration	
complete	Instrument		4.2.1.1.8	Phase A		System shall be able to display all the instruments (typeA, typeB, type C, typeD) based on the location and method.	Must	Configuration	
complete	Sample		4.2.1.1.9	Phase A		System shall be able to select instrument by scanning.		Standard	
complete	LES		4.2.1.1.10	Phase B		System shall be able to select multipe instruments in LES.	Must	Standard	
complete	LES		4.2.1.1.11	Phase B		System shall be able to have option to manually override for the parameters captured.	Must	Standard	
complete	Assignment		4.2.1.1.12	Phase B		System shall be able to assign test based on the versatility matrix from OSDP system once integrated.		Customisation	
complete	Assignment		4.2.1.1.13	Phase B		System shall be able to assign test based on the sample status (Waiting to start).		Customisation	
complete	Assignment		4.2.1.1.14	Phase B		System shall be able to auto assign for the current date based on the schedule configured.		Customisation	
complete	Assignment		4.2.1.1.15	Phase B		System shall be able to monitor samples on a certain interval basis and assign it.		Customisation	
complete	Assignment		4.2.1.1.16	Phase A		System shall be able to assign samples manually.	Must	Customisation	
complete	Assignment		4.2.1.1.17	Phase B		System shall be able to assign samples based on SAP test skip lot rules.		Customisation	
complete	Assignment		4.2.1.1.18	Phase B		System shall be able to assign samples based on other pending QC works.		Customisation	
complete	Assignment		4.2.1.1.19		mark	Henkel team send us logical rule about the assign task automatically.			
complete	Assignment		4.2.1.1.20		mark	Henkel team send us the detail information thermofisher listed of OSDP. Noted: Sync daily. (TF list the detail information (mapping fileds required) first)			
complete	Sample		4.2.1.1.21	Phase A		System shall be able to maintain master data which mapping analysis and instrument type.	Must	Standard	
complete			4.2.1.2			The system should allow Lab supervisor to determine sample test schedule arrangement way. And the system is capable of allow QC supervisor to stop automatic schedule and change it to manually.	Must		
complete	Assignment		4.2.1.2.1	Phase B		System shall be able to modify schedule.	Must	Customisation	
complete			4.2.1.3			The system should have capability to allow QC Supervisor modify testing task sequence based on lab test priority.	Must		
complete	Assignment		4.2.1.3.1	Phase B		System shall be able to allow key user modify task based on urgency test.	Must	Configuration	
complete			4.2.1.4			Standard testing Pre-schedule: The system should have capability to get SAP inspection lot immediately after release in SAP and prearrange testing schedule based on inspection lot sequence.	Must		

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Document Ref: Appendix 01_Functional Requirements_LIMS - Thermo Fisher
Version: 1
Date: 04 June 2024

gathering status	Module	Req. No.	URS Ref.	Delivery Phase	Functional Area / Workshop	Requirement	Priority	Type	Estimate
complete	Interface		4.2.1.4.1	Phase A		system shall be able to create tasks when SAP pushes inspection lot based on configurable interval and configurable start time.		Configuration	
complete	Assignment		4.2.1.4.2	Phase B		System shall be follow FIFO rules based on sample receiving date when assigning tasks.		Customisation	
complete	Interface		4.2.1.4.3	Phase A		System shall assign tasks before the inspection lot only for JIT materials. And the results shall be pushed to sap if inspection lot is created.		Customisation	
Complete			4.2.1.4.4		mark	frequency need to double confirm by TF and Henkel; to be confirmed.			
complete			4.2.1.5			Standard testing schedule: The system should have capability to determine inspection task sequence based on preschedule once the sample arrives.	Must		
complete	Assignment		4.2.1.5.1	Phase B		System shall assign task based on priority rules once sample arrives.		Customisation	
complete			4.2.1.6			Non-standard testing/non-label sample Pre-schedule: The system should be capable of prearrange test schedule based on test request sequence or based on calendar.	Must		
complete	Assignment		4.2.1.6.1	Phase B		System shall allow to create tasks based on calendar schedule.		Customisation	
complete			4.2.1.7			Non-standard testing/non-label sample schedule: The system should be capable of determine testing schedule. Check for sample location from QCLAD, if already available use the label in the sample and start testing process. In case, sample not available at QC	Must		
complete	Report&Label		4.2.1.7.1	Phase A		System shall allow user to print label by manually input the information.	Must	Configuration	
complete			4.2.1.8			System should be capable to maintain QC shift, user information to support scheduler feature. Number of shift and shift hour of each shift can be configurable by Lab supervisor.	Must		
complete	Assignment		4.2.1.8.1	Phase A		System shall allow user maintain master data for shift schedule by week. Number of shift and shift hour of each shift shall be configurable in the master data at plant level.		Customisation	
complete			4.2.1.8.2		mark	note: Henkel need offer shift template			
complete			4.2.1.9			The system should be capable to get/check user testing skill via Henkel’ s skill matrix system and assign the right person. In case if skill matrix not available or the supervisor wants to modify QCLAD’ s task assignment system should allow manual assignment.	Must		

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Document Ref: Appendix 01_Functional Requirements_LIMS - Thermo Fisher
Version: 1
Date: 04 June 2024

gathering status	Module	Req. No.	URS Ref.	Delivery Phase	Functional Area / Workshop	Requirement	Priority	Type	Estimate
complete	Assignment		4.2.1.9.1	Phase B		System shall check skill when assigning tasks to operator.		Customisation	
complete	Interface		4.2.1.9.2	Phase B		System shall be able to support skill shall be sync with OSDP system if there is OSDP in plant. Refer to 4.5.1.14		Customisation	
complete	Assignment		4.2.1.9.3	Phase A		System shall allow user to rearrange task manually.	Must	Configuration	
complete			4.2.1.10			For standard and non-standard testing, test status should be showed on dashboard, the detailed information, including the inspection lot(standard test), test request#(non-standard test), product number, product name, batch#, user ID, etc.	Must		
complete	Dashboard		4.2.1.10.1	Phase A		System shall have dashboard which display tests, test status, product number, product name, batch number, user ID. inspection lot, inspection lot(standard		Customisation	
complete			4.2.1.10.2		mark	Henkel offer dashboard template, Tony			
complete			4.2.1.11			The system should be capable of arrange test schedule for scheduled tests (shelf life study, aging test, etc.) based on test schedule @ specific frequency as monthly, weekly, specific time on every day	Must		
complete	Interface		4.2.1.11.1	Phase B		System shall create tasks for samples based on specific frequency.		Customisation	
			4.3.1.1		COA of RM auto-capture	The system shall provide functionality to recognize & download documents from incoming e-mail	Wont	Out Of Scope	
			4.3.1.1.1						
			4.3.1.2			The system shall provide functionality to extract data from COA and store them in Henkel database	Wont	Out Of Scope	
			4.3.1.2.1						
			4.3.1.3			The system shall provide functionality to check open SAP inspection lot and search respective data in database	Wont	Out Of Scope	
			4.3.1.3.1						
			4.3.1.4			The system shall provide functionality to validate characteristics, results, and range (COA against SAP)	Wont	Out Of Scope	
			4.3.1.4.1						
			4.3.1.5			The system shall provide functionality to record result and transfer,	Wont	Out Of Scope	
			4.3.1.5.1						
			4.3.1.6			The system shall provide functionality to approach QC through email when abnormal happened or wrong data generated	Wont	Out Of Scope	
			4.3.1.6.1						
			4.3.1.7			Henkel has several language and format based on vendor, the system should have capability to capture from any format.	Wont	Out Of Scope	

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gathering status	Module	Req. No.	URS Ref.	Delivery Phase	Functional Area / Workshop	Requirement	Priority	Type	Estimate
			4.3.1.7.1						
complete			4.3.2.1		Data Capture of Testing Instrument	The system shall provide functionality to capture data from Henkel type A and B test instrument automatically	Must		
complete	IM Integartion		4.3.2.1.1	Phase A		System shall be able to capture data from Henkel type A.		Interface	
complete	IM Integartion		4.3.2.1.2	Phase A		System shall be able to capture data from Henkel type B.		Interface	
complete			4.3.2.2			The system shall provide functionality to input the testing data manually	Must		
complete	Sample		4.3.2.2.1	Phase A		System shall be able support to input the testing data manually.		Standard	
complete			4.3.2.3			The system shall provide functionality to integrate all the types and brands of instruments, simplifying and streamlining the operation	Must		
complete	LES		4.3.2.3.1	Phase B		System shall be able to select and display the instrument in the LES.		Standard	
complete	IM Integartion		4.3.2.3.2	Phase A		System shall provide functionality to integrate all type A and type B instruments which can export readable file.		Configuration	
complete	LES		4.3.2.3.3	Phase B		System shall be able to fetch the results in LES and display the results.		Standard	
complete			4.3.2.4			The system shall provide functionality to automatically identify test sample information via test instruments without additional manually key in sample information into test instruments.	Must		
complete			4.3.2.4.1	Phase B	mark	user need to identifying the sample id in instrument software so that LIMS can match data to samples.			
complete	LES		4.3.2.4.2	Phase B		System shall be able to identify sample in LES by scan label.		Configuration	
complete	LES		4.3.2.4.3	Phase B		System shall be able to manually enter sample id in LES.		Configuration	
complete			4.3.2.5			The system vendor should train Henkel user about how to set up functionality to capture instrument data	Must		
complete	IM Integartion		4.3.2.5.1	Phase A		System vendor should train Henkel user about how to set up functionality to capture instrument data.		Non-function	
complete			4.3.2.6			The system vendor has overall responsibility to check Henkel instrument connection capability and set up interface between instrument and system.	Must		
complete	IM Integartion		4.3.2.6.1	Phase A		System vendor has overall responsibility to check Henkel instrument connection capability and set up interface between instrument and system.		Non-function	

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Document Ref: Appendix 01_Functional Requirements_LIMS - Thermo Fisher
Version: 1
Date: 04 June 2024

gathering status	Module	Req. No.	URS Ref.	Delivery Phase	Functional Area / Workshop	Requirement	Priority	Type	Estimate
complete			4.4.1.1		Laboratory Execution	Enable Workflow execution of Laboratory daily operation Quick overview of inspection plan on daily, weekly, monthly views Able to view the status of testing, details of inspection lot on clicking on each On clicking user is expected to see the status of SOP (which steps completed, on-going and not started) Kanban dashboards with storage location and quantity (with/within safety stock) with desired filters Dashboards showing lab equipment availability	Must		
complete	Dashboard		4.4.1.1.1	Phase A		System shall support dashboard display overview of inspection lot on daily, weekly, monthly views.		Customisation	
complete	Sample		4.4.1.1.2	Phase A		System shall be able to view the status of testing, details of inspection lot.		Configuration	
complete	Sample		4.4.1.1.3	Phase A		System shall be able to record standard cycle time and actual cycle time under test level.		Customisation	
complete	LES		4.4.1.1.4	Phase B		System shall be able to dispaly the status of LES (which steps completed, in progress and not started).		Standard	
complete	Dashboard		4.4.1.1.5	Phase A		System shall be able to display sample storage location and sample quantity.		Customisation	
complete	Dashboard		4.4.1.1.6	Phase A		System shall support dashboard showing lab equipment availability. Equipment usage can be tracked trhough scanning before and after the usage.		Customisation	
complete			4.4.1.2			Break each SOP down into steps that lead through the process, including test method execution, instrument calibration, result check and calculation Henkel has different test methods, the system should have capablity to support all test method execution. Henkel test method instruction keeps in Henkel Remap system. The system should have a hyperlink to guide user to Remap system on clicking a test method. Master hyperlink guide user to Remap system. The system should allow site admin to maintain unique hyperlink for each different test method.	Must		
complete			4.4.1.2.1		mark	Henkel need to offer 10 feasible SOP so that TF team can creat 10 LES in system.			
complete	LES		4.4.1.2.2	Phase B		System shall support maintain SOP by steps in LES, including test method execution, instrument calibration, result check and calculation.		Standard	
complete	LES		4.4.1.2.4	Phase B		System shall allow user to access Remap system by hyperlink in analysis.		Configuration	
complete	LES		4.4.1.2.4	Phase B		System shall support all test method execution.		Standard	
complete	LES		4.4.1.2.5	Phase B		System shall allow specific user to mantain and view hyperlink in analysis.		Configuration	
complete			4.4.1.3			Step-wise approach to achieve Automation of routine tasks User should be able to start and record testing results from iPAD / Tablet Ability to upload documents during testing(like images..)	Must		
complete	LES		4.4.1.3.1	Phase B		System shall support user to start and record testing results by step from iPAD or Tablet.		Standard	
complete	LES		4.4.1.3.2	Phase B		System shall be able to upload attachment during testing(like images..)		Standard	
complete			4.4.1.4			Enable Instrument Integration during execution of laboratory SOP. Provides the ability to retrieve values from remotely attached instruments	Must		

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Document Ref: Appendix 01_Functional Requirements_LIMS - Thermo Fisher
Version: 1
Date: 04 June 2024

gathering status	Module	Req. No.	URS Ref.	Delivery Phase	Functional Area / Workshop	Requirement	Priority	Type	Estimate
complete	LES		4.4.1.4.1	Phase B		System shall support retrieve values from Type A and Type B instruments in LES.		Configuration	
complete			4.4.1.4.1		MARK	remotely attach instrument = type A and type B instrument			
complete			4.4.1.5			Enable eSignatures and Cross Check between different users during execution	Must		
complete	LES		4.4.1.5.1	Phase B		System shall support Cross Check between different users during execution in LES.		Standard	
complete			4.4.1.5.2		mark	what's e signature(like peer review)stands for? Re: Esignature is out of scope. Replace eSigniture with peer review.			
complete			4.4.1.6			Enable audit trail on all steps during execution	Must		
complete	LES		4.4.1.6.1	Phase B		System shall support audit on all steps during execution.		Standard	
complete			4.4.1.7			Allows execution of Lab Methods in remote locations, can be used in PC, tablet(Ipad, etc.)	Must		
complete	LES		4.4.1.7.1	Phase B		System shall support execution of Lab Methods in remote locations, can be used in PC, tablet(Ipad, etc.)		Standard	
complete			4.4.1.8			Review the actual Lab Execution to ensure it has been executed correctly	Must		
complete	LES		4.4.1.8.1	Phase B		System shall support review Lab Execution.		Standard	
complete			4.4.1.9			Generate follow-up actions from the Lab Execution review	Must		
complete	LES		4.4.1.9.1	Phase B	mark	System shall allow specific user to reassign the test to another user.		Configuration	
complete	LES		4.4.1.9.2	Phase B		System shall require to enter comments when rejecting the result during Lab execution review.		Configuration	
complete			4.4.1.10			Re-generate the paper SOP from the configured Lab methods, including any execution outcomes, Access to all Lab Execution data for reporting and Display audit information, linked entities, review statuses	Must		
complete	LES		4.4.1.10.1	Phase B		System shall support print Lab execution record.		Standard	
complete	LES		4.4.1.10.2	Phase B		System shall support configure lab execution record report template.		Standard	
complete			4.4.1.11			Enable workflow to execute Henkel MSA, maintain MSA plan, send reminder according to plan, send alerts once expired MSA happened.	Must		
complete	Sample		4.4.1.11.1	Phase B		System shall support execute and maintain MSA plan.		Configuration	
complete	Sample		4.4.1.11.2	Phase B		System shall support remind MSA according to plan and remind expiration.		Configuration	
complete			4.4.2.1		Resampling	Four solutionsare possible to handle the resampling: •Create a new inspection lot in SAP •Create a new inspection lot in SAP with a link to the initial inspection lot, for example a user field populated with the initial lot. •Create another sample in the QCLAD, related to the inspection lot (job in the QCLAD) Reactivate in the QCLAD a validated sample and reenter the results on the initial test. Then the sample is authorised and all the results are sent again to SAP. The result values are updated in SAP	Must		
pending	Interface		4.4.2.1.1	Phase B	Mark	Create another sample in the QCLAD, related to the inspection lot (job in the QCLAD) and all the results are sent again to SAP. The result values are updated in SAP. Check with Neelima System shall be able to trigger resample in QCLAD and create a new sample with unique ID, which results can be sent back to SAP.			

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Document Ref: Appendix 01_Functional Requirements_LIMS - Thermo Fisher
Version: 1
Date: 04 June 2024

gathering status	Module	Req. No.	URS Ref.	Delivery Phase	Functional Area / Workshop	Requirement	Priority	Type	Estimate
complete			4.4.2.1.2		Mark	Henkel need to check use which one solution for handling resampling.			
complete			4.5.1		Interface to Smart factory, MES and SKill matrix	Enable Download of information from SAP to QCLAD and upload information from QCLAD to SAP Push/pull information from Henkel smart factory, MES, Henkel Skill Matrix. The information pulled/pushed should be available in QCLAD in case of analyzing. API calls need to be designed and agreed by different teams for pushing/pulling data based on the project scope.	Must		
complete	Interface		4.5.1.1	Phase A		System shall support download inspection lot from SAP to QCLAD. 1) Create a sample in QCLAD based on the downloaded Inspection Lot, or change an existing sample (JIT process). 2) If the inspection lot in SAP also has multiple activities, then QCLAD needs to create multiple samples. 3) The test item in lot has a field for whether or not to skip the batch (dynamic message), and whether the SMIDI standard function can be written directly to the test table.		Interface	
complete	Interface		4.5.1.2	Phase A	Mark	System shall support download Batch information from SAP to QCLAD.		Interface	
			4.5.1.3			System shall support download Inspection Plan from SAP to QCLAD. Needs to be synchronised and updated in real time.		Interface	
complete	Interface		4.5.1.4	Phase A		The system shall support the downloading of Inspection plans from SAP to QCLAD whose Inspection Plan contains the MIC. Needs to be synchronised and updated in real time.		Interface	
complete	Interface		4.5.1.5	Phase A		The system shall support the downloading of Inspection plans from SAP to QCLAD whose Inspection Plan contains the specification. Needs to be synchronised and updated in real time.		Interface	
complete	Interface		4.5.1.6	Phase A		The system shall support the downloading of Inspection plans from SAP to QCLAD whose Inspection Plan contains the Material master data. Needs to be synchronised and updated in real time.		Interface	
			4.5.1.7	Phase A		System shall support download GHS info from SAP to QCLAD. Refer to 4.8.5.8.2		Out Of Scope	
			4.5.1.8	Phase A		System shall support download Plant Code from SAP to QCLAD.		Interface	
complete	Interface		4.5.1.9	Phase A		System shall support download QM notification from SAP to QCLAD.		Interface	
complete	Interface		4.5.1.10	Phase A		System shall support to upload result value from QCLAD to SAP. 1) When there is a Retest operation done, the result of the last Retest needs to be returned to SAP. 2) once the LIMS post result back to SAP, SAP MIC status should be auto rejected or accepted? 3) system shall support post all the samples result back to SAP, including the resampled sample. 4) related to item 1, whether Smidi can configure that, once inspection lot is downloaded mapped with an existing sample in LIMS; if that sample is authorized with result, then Smidi can also trigger the result pushing to SAP? (JIT process)		Interface	
complete	Interface		4.5.1.11	Phase A	Mark	System shall support to upload result value from SAP to QCLAD.		Out Of Scope	
complete	Interface		4.5.1.12	Phase B		System shall push notify information(including resample and adjustment) to Henkel smart factory.		Interface	
complete	Interface		4.5.1.13	Phase B		System shall pull result information from Henkel smart factory.		Interface	
complete	Interface		4.5.1.14	Phase B		System shall support to upload result value from QCLAD to Henkel smart factory.		Interface	
complete	Interface		4.5.1.15	Phase B		System shall support to upload sample states from QCLAD to Henkel smart factory.		Interface	
complete	Interface		4.5.1.16	Phase B		System shall push/pull adjustment quantity information from/to Henkel smart factory.		Interface	
complete	Interface		4.5.1.17	Phase B		System shall pull operator skill information from Henkel Skill Matrix.		Interface	

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Document Ref: Appendix 01_Functional Requirements_LIMS - Thermo Fisher
Version: 1
Date: 04 June 2024

gathering status	Module	Req. No.	URS Ref.	Delivery Phase	Functional Area / Workshop	Requirement	Priority	Type	Estimate
complete			4.5.2		SAP process in existing applications	Currently MES and Smart factory have SAP interface to pull master and transactional data. Need to identify what to be pulled from SAP directly and can we reuse the data pulled by the applications to SAP instead of pulling again. Need to work out on the details.	Must		
complete	Interface		4.5.2.1	Phase A		Refer to 4.5.1.1 & 4.5.1.2 & 4.5.1.3 above.		Interface	
complete			4.5.3		Communication between SAP and QCLAD	nformation downloaded : •IDH#, Batch#, master data (plant#, lab#, material, shelf life, date of manufacturing, inspection interval, MIC#, etc.), inspection plan, skip lot rules, Quality Notification#, equipment maintenance master data •Details of required fields to be agreed between Henkel and vendor after the process agreement. Please don't consider this as a final set of	Must		
complete	Interface		4.5.3.1			Refer to 4.5.1.1 & 4.5.1.2 & 4.5.1.3 above.		Interface	
complete			4.5.4		Upload of information from the QCLAD to SAP/MES/Smart factory	The upload of the results is triggered by the validation of the laboratory. Once user approves the results pulled from lab equipment in QCLAD, it should be pushed to SAP in near real time. In-process testing result need to refer in-process testing workflow to update test result and decision into Henkel MES and smart factory system. If the laboratory reactivates the sample and the results, a result value can be sent several times to SAP, the result in SAP is updated (the characteristics are declared as long term characteristics in SAP).	Must		
complete	Sample		4.5.4.1			System shall allow result values can be modified with a pre-defined rules or other calculations automatically.		Standard	
complete	Sample		4.5.4.2			System shall support the specification limit can be setup in the QCLAD and is associated to each result.		Standard	
complete			4.6.1.1		Dashboard / Report (downloadable)	Sample status summary graph per Lab/Plant/Country/Henkel network/ global Sample status graph can be reviewed daily/weekly/monthly/yearly Sample status should have completed, in progress, received, cancelled Report should include all sample type, RM, semi-finish, Bulk, FG, standard test and non-standard test, etc. Report can be filtered by material, technology (product technology information should from SAP)	Must		
complete	Dashboard		4.6.1.1.1	Phase A		System shall be able to display sample status by lab/plant/country/henkel network/global in dashboard.		Customisation	
complete	Dashboard		4.6.1.1.2	Phase A		System shall be able to display sample status by daily/weekly/monthly/yearly in dashboard.		Customisation	
complete	Dashboard		4.6.1.1.3	Phase A		System shall support different sample status, including completed, in progress, received, cancelled, retained.		Customisation	
complete	Dashboard		4.6.1.1.4	Phase A		System shall include sample type in report, detailed sample type refers to RTM 4.5.1 items.		Customisation	
complete	Sample		4.6.1.1.5	Phase A		System shall support report can be filtered by material, technology (product technology information should from SAP) in folder.		Configuration	
complete					Mark	Henkel need to check what's the meaning of report with Neelima			
complete			4.6.1.2			Daily sample status summary graph per Lab/Plant should be showed real time.	Must		
complete	Dashboard		4.6.1.2.1	Phase A		System shall support to display dashboard with sample status in real time.		Customisation	
complete			4.6.1.3			Scheduled job list (Task assignment job in specific time period) Scheduled job list is related to schedule work. Scheduled job list should be showed real time.	Must		
complete	Dashboard		4.6.1.3.1	Phase A		System shall display task list in specific time period.		Customisation	
complete	Dashboard		4.6.1.3.2	Phase A		System shall display task list related to work.		Customisation	
complete	Dashboard		4.6.1.3.3	Phase A		System shall display task list in real time.		Customisation	

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Document Ref: Appendix 01_Functional Requirements_LIMS - Thermo Fisher
Version: 1
Date: 04 June 2024

gathering status	Module	Req. No.	URS Ref.	Delivery Phase	Functional Area / Workshop	Requirement	Priority	Type	Estimate
complete			4.6.1.4			Overdue scheduled job report	Must		
complete	Dashboard		4.6.1.4.1	Phase A		System shall display overdue task list.		Customisation	
complete			4.6.1.5			Instrument status (available, out of service, Retired) Statistics (key: instrument ID, instrument Type, days in available, days out of service…)	Must		
complete	Dashboard		4.6.1.5.1	Phase A		System shall display instrument list with status(Unaviable, Available, Maintance Due, Retired, Require Calibration, In Calibration).		Standard	
complete	Dashboard		4.6.1.5.2	Phase A		System shall support dashbord to display instrument list including (instrument ID, instrument Type, instrument Status, days in available and days out of service by daily, weekly, monthly or yearly)		Customisation	
complete			4.6.1.5.3		Mark	What’ s the meaning of ’days in available, days out of service’? 怎么理解Days			
complete			4.6.1.6			Instrument load and utilization, Actual Instrument Utilization=Σ(Total Actual Instrument working Hours/ Total working Hours) Total working hours is related to shift arrangement.	Must		
complete	Dashboard		4.6.1.6.1	Phase A		System shall support display instrument utilization by daily/weekly/monthly/yearly in dashboard.		Customisation	
complete	Dashboard		4.6.1.6.2	Phase A		Total Actual Instrument working Hours= End time by scanning minus Start time by scanning		Customisation	
complete	Dashboard		4.6.1.6.3	Phase A		Total working hours is related to shift arrangement,calculated as off-duty time minus on-duty time.		Customisation	
complete			4.6.1.7			Work load (Sample or analysis) statistics (per work hour, per plant, per analyst…)	Must		
complete	Dashboard		4.6.1.7.1	Phase A		System shall support display work load by plant/duration.		Customisation	
complete			4.6.1.8			Work load can be number of inspection lot, number of sample, number of test, number of skip lot, number of skip test	Must		
complete	Dashboard		4.6.1.8.1	Phase A		System shall support display work load with number of inspection lot, number of sample, number of test, number of skip lot, number of skip test in folder.		Customisation	
complete			4.6.1.9			Report: extract data based on sample type, analysis, test schedule, login time, standard time….	Must		
complete	Sample		4.6.1.9.1	Phase A		System shall support export data based on sample type, analysis, test schedule, login time, standard time by folder.		Standard	
complete			4.6.1.9.2		Mark	what format of the report? The aim of extract data.			
complete			4.6.1.10			QC release time: To be able to automatically calculate individual batch/test request complete time in specific time period QC Release Time=Σ(Time End-Time Start) Time End: UD time of batch or test request completed time Time Start: Sample received time (actual time barcode scan in QC) of inspection lot	Must		
complete	Dashboard		4.6.1.10.1	Phase A		System shall be able to automatically calculate QC release time for individual batch/test request.		Customisation	
complete	Dashboard		4.6.1.10.2	Phase A		Sample Release Time=Time End-Time Start		Customisation	
complete			4.6.1.10.3		Mark	Why need Σ? Henkel:remove Σ			
complete			4.6.1.10.4		Mark	Time End: sample authorizd time			
complete	Dashboard		4.6.1.10.5	Phase A		Time Start: sample received time (actual time barcode scan in QC) of inspection lot		Customisation	

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Document Ref: Appendix 01_Functional Requirements_LIMS - Thermo Fisher
Version: 1
Date: 04 June 2024

gathering status	Module	Req. No.	URS Ref.	Delivery Phase	Functional Area / Workshop	Requirement	Priority	Type	Estimate
			4.6.1.11			QC Productivity: Measure efficiency of QC efforts at a manufacturing site in specific time period 1. QC Testing Productivity by test =∑(# test)/#FTE The sum of # inspection test compare with #FTE for a give selection 2. QC Testing productivity by inspection lot =∑(# SAP inspection lot)/#FTE	Must		
pending	Dashboard		4.6.1.11.1		Mark	Henkel need to check what's the report of Productivity Henkel:QC productivity can be valuated according to 3 formulas: QC efficiency 1=Direct QC test LH(IQC+IPQC+OQC,including retest)/QC attendence hours QC efficiency 2=(Direct QC test LH+non-standard QC test LH)/QC attendence hours QC efficiency 3=(Direct QC test LH+non-standard QC test LH+indirect QC LH)/QC attendence hours Standard testing - Direct QC Test Non standard testing - Indirect QC QC attendance hours - based on the shift management		Customisation	
complete			4.6.1.12			Instrument PM(preventive maintenance) performance, including, number of PM performed in QC Lab & on time PM% in specific time period	Must		
complete	Instrument		4.6.1.12.1	Phase A		System shall allow user to add maintenance plan. Maintenance plan shall include PM(preventive maintenance) type.		Configuration	
complete	Dashboard		4.6.1.12.2	Phase A		System shall display on time proportion for PM for each instrument by report.		Customisation	
complete			4.6.1.13			Instrument calibration performance: including, number of calibration performed in QC Lab & on time calibration% in specific time period	Must		
complete	Dashboard		4.6.1.13.1	Phase A		System shall display the quantity of calibration record and on time proporation for maintenance by report.		Customisation	
complete			4.6.1.14			MSA performance: including, number of MSA performed in QC Lab & on time MSA% in specific time period	Must		
complete	Sample		4.6.1.14.1	Phase A		System shall display the quantity of MSA task completed in system, completed shall be triggered by uploading MSA report to QCLAD.		Configuration	
complete	Dashboard			Phase A		System shall display on time proportion for MSA task by report.		Customisation	
complete			4.6.1.15			Consumable/Spare parts report: stock status and safety stock status in specific time period	Must		
complete	Dashboard		4.6.1.15.1	Phase A		System shall display stock status and safety stock status(the stock is below the minumum stock and above the maximum stock) in specific time period by report.		Customisation	
complete			4.6.1.15.2		Mark	status refers to what?			
complete			4.6.1.16			Retain sample report: Analysis by storage location, stock status , number of expire sample stock status, disposed stock by date	Must		
complete	Dashboard		4.6.1.16.1	Phase A		System shall display retain sample in dashboard by storage location, retain status , number of expired retains, disposed retains by date.		Customisation	
			4.6.1.16.2		Mark	need dashboard. Henkel : design the dashborad			
complete			4.7.1.1		Sample Tracking	The system shall maintain a history of sample life information about a sample's workflow. The workflow is a requirement for internal COC (Chain of Custody) that tracks all samples from the time they are received until disposal	Must		
complete	Sample		4.7.1.1.1	Phase A		System shall support to track sample life history (Chain of Custody).	Must	Configuration	
complete	Sample		4.7.1.1.2	Phase A		System shall support to display samples and test results by batch number.		Standard	

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Document Ref: Appendix 01_Functional Requirements_LIMS - Thermo Fisher
Version: 1
Date: 04 June 2024

gathering status	Module	Req. No.	URS Ref.	Delivery Phase	Functional Area / Workshop	Requirement	Priority	Type	Estimate
complete			4.7.1.2			The system shall track users by log-in the data they enter and functions they perform	Must		
complete	Security		4.7.1.2.1	Phase A		System shall support to track data entry history and modification history.		Configuration	
complete			4.7.1.3			The user shall be able to view all the information in QCLaD dashboard about a sample including current status	Must		
complete	Sample		4.7.1.3.1	Phase A		The user shall be able to view all the information about a sample including current status in QCLaD folder.		Standard	
complete			4.7.1.3.2		Mark	Noted: can fulfill by folder configuration.			
complete			4.7.1.4			The user shall be able to update the sample’ s workflow information as it moves through the process	Must		
complete			4.7.1.4.1		Mark	does ’update the sample’ s workflow information’ means modify samples ’s attributes ? Dragon -- No, attributes is not been changed but the status may change. Similar to No.36			
complete	Sample		4.7.1.4.2	Phase A		System shall update sample status automatically according to workflow.		Standard	
complete			4.7.1.5			The system shall track the current status of test(s) for a sample	Must		
complete	Sample		4.7.1.5.1	Phase A		System shall supoort to display the current status of test(s) for a sample.		Standard	
complete			4.7.1.6			The system shall display a history of a sample’s workflow	Must		
complete	Security		4.7.1.6.1	Phase A		System shall support to track a history of a sample lifecycle.		Configuration	
complete			4.7.1.6.2		MARK	does workflow means lifecycle? Re: Lifecycle include multiple workflow			
complete			4.7.2.1		Data View and Audit Trail	The system shall be capable of supporting multiple languages(based on the plant preferred local language) and English, to support Henkel global roll out	Must		
complete	Sample		4.7.2.1.1	Phase A		System shall support multiple languages. (Chinese and English will be implemented in Dragon pilot phase.)		Customisation	
complete			4.7.2.2			The system shall provide functionality for users with appropriate permissions to configure a view of the QCLAD data; Permission can be configurable, by function or by user	Must		
complete	Security		4.7.2.2.1	Phase A		System shall support data access control by function department or by user role.		Standard	
complete	Security		4.7.2.2.2	Phase A		System shall support to configure user’s role and group.		Standard	
complete			4.7.2.3			The system shall provide real time electronic sample status and location tracking (Lab location and area).	Must		
complete	Sample		4.7.2.3.1	Phase A		System shall support to display sample status and location in real time.		Standard	
complete			4.7.2.4			The system shall provide real time access to results for peer and supervisor review during sample testing (electronic / hard copy), hardcopy can be download/print from system	Must		
complete	Sample		4.7.2.4.1	Phase A		system shall support to display results for peer and supervisor review in real time.		Standard	
complete	Sample		4.7.2.4.2	Phase A		hard copy print requirement refers to RTM 4.4.1.10.1.		Standard	
complete			4.7.2.5			The system shall provide functionality to perform trending and simple data analysis	Must		

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Document Ref: Appendix 01_Functional Requirements_LIMS - Thermo Fisher
Version: 1
Date: 04 June 2024

gathering status	Module	Req. No.	URS Ref.	Delivery Phase	Functional Area / Workshop	Requirement	Priority	Type	Estimate
complete			4.7.2.5.1		MARK	Thermo SQC function instruction. Re:TF have upload instruction in SharePoint and replied it via email. Re:Henkel replied Not in the scope of QC LaD		Out Of Scope	
complete			4.7.2.6			The system shall allow data generated to be exported in XML / CSV/ PDF format	Must		
complete	Sample		4.7.2.6.1	Phase A		system shall support to export data through folder explorer in XML / CSV/ PDF format.		Standard	
complete			4.7.2.7			The system shall provide functionality to export data in a usable format to third party software (e.g. Excel)	Must		
complete	Sample		4.7.2.7.1	Phase A		System shall support to export data through folder explorer in an usable format.		Standard	
complete			4.7.2.7.2		MARK	what's third party software? Re: We can download data from QCLaD to excel,pdf,word. We can upload data of third party software as a excel/pdf/word format to QCLaD manually.			
complete			4.7.2.8			The system shall provide a simple interface / method for extraction of data into statistic report	Must		
complete	Sample		4.7.2.8.1	Phase A		System shall provide a simple template or method for extraction of data.		Standard	
complete			4.7.2.8.2		MARK	interface means integration? Why need interface in here. Re:Template			
complete			4.7.2.9			The system shall provide tools to search for datasets or components by any combination of fields	Must		
complete	Sample		4.7.2.9.1	Phase A		System shall support to search for datasets or components by combination of fields in folder explorer.		Standard	
complete			4.7.2.10			The system should provide tools for users to save one or more defined sets of structured search parameters for future use, linked to the user's profile	Must		
complete	Sample		4.7.2.10.1	Phase A		System shall support user to create personal folder by saved defined one or more criteria to display certain data.		Standard	
complete			4.7.2.11			The system shall provide full audit trails. All audit trails shall be secure, computer generated, time stamped using the local time. It shall include the date and time, retention of original data, reason for change to original data and the individual performing the task. All data entry, recording of reporting processes and all changes to the reported data shall be recorded along with any deletion of data	Must		
complete	Security		4.7.2.11.1	Phase A		System shall provide full audit trails.		Standard	
complete	Security		4.7.2.11.2	Phase A		System shall support all audit trails shall be secure, computer generated, time stamped using the local time.		Standard	

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Document Ref: Appendix 01_Functional Requirements_LIMS - Thermo Fisher
Version: 1
Date: 04 June 2024

gathering status	Module	Req. No.	URS Ref.	Delivery Phase	Functional Area / Workshop	Requirement	Priority	Type	Estimate
complete			4.7.2.11.3		Mark	Thermo check local time Re: have checked.			
complete	Security		4.7.2.11.4	Phase A		System shall support audit trial shall include the date and time, retention of original data, reason for change to original data and the individual		Standard	
complete	Security		4.7.2.11.5	Phase A		System shall be able to support all data entry, recording of reporting processes and all changes to the reported data shall be recorded along with any deletion of data.		Standard	
complete			4.7.2.12			The system shall require audit trail entries for all creation, modification and approval associated with static data	Must		
complete	Security		4.7.2.12.1	Phase A		System shall support audit trail for all creation, modification and approval associated with static data.		Standard	
complete			4.7.2.13			The system shall allow the display, print or download any / all audit trails	Must		
complete	Security		4.7.2.13.1	Phase A		System shall support to display, print or export any / all audit trails.		Standard	
complete			4.7.2.14			The system should be able to export the audit trail into a report	Must		
complete	Security		4.7.2.14.1	Phase A		System shall support to export the audit trail into a report.		Standard	
complete			4.8.1.1		Laboratory equipment	Standard management of instruments and equipment in the lab including specifications, parameters, spare parts, calibration and maintenance.	Must		
complete	Instrument		4.8.1.1.1	Phase A		System shall support maintain specification and parameters in equipment master data.		Standard	
complete	Instrument		4.8.1.1.2	Phase A		System shall allow maintain spare parts, calibration and maintenance.		Standard	
complete			4.8.1.2			Instrument calibration and maintenance scheduler. A module to maintain calibration schedules and maintenance schedule and providing warning messages. Use instrument calibration module.	Must		
complete	Instrument		4.8.1.2.1	Phase A		System shall suport calibration and maintenance plan locally.		Standard	
complete	Instrument		4.8.1.2.2	Phase A		System shall remind user when instrument maintenance expire in folder.		Configuration	
complete			4.8.1.2.3		Mark	Need to check maintenance scheduler is in scope with Neelima. Henkel:Need the maintainance task to be schedulered, but sending notification to SAP is not in scope.			
complete			4.8.1.3			Instrument history. An instrument history report detailing the failures and calibration / maintenance events for an instrument	Must		
complete	Instrument		4.8.1.3.1	Phase A		System shall support to record instrument history , including failures, calibration, maintenance.		Standard	
complete			4.8.1.4			When the chemical/instrument is going to be expired, the system shall provide functionality to alert specified user(s) accordingly	Must		
			4.8.1.4.1			System shall alert specific user when instrument is about to expire list in folder.		Out Of Scope	

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Document Ref: Appendix 01_Functional Requirements_LIMS - Thermo Fisher
Version: 1
Date: 04 June 2024

gathering status	Module	Req. No.	URS Ref.	Delivery Phase	Functional Area / Workshop	Requirement	Priority	Type	Estimate
complete			4.8.1.5			Result entry can be blocked for users that do not comply with the user testing skill requirements, QCLaD should has capability to check user' s testing skill from Henkel skill matrix tool if Henkel site use the tool	Must		
complete	Sample		4.8.1.5.1	Phase A		System shall only allow tests assign to qualified users.		Configuration	
complete	Interface		4.8.1.5.2	Phase B		System shall sync user' s testing skill from Henkel skill matrix tool.		Interface	
complete			4.8.1.6			The system shall provide a way to alert the system defined users when the chemicals or instruments used for the test is/are expired Exception for expired calibration/maintenance can be managed by NCMR, enter NCMR number can continue use the instrument.	Must		
complete	Stock		4.8.1.6.1	Phase A		System shall alert specific user when chemicals or instruments is/are expired list in folder.		Configuration	
complete	Instrument		4.8.1.6.2	Phase A		System shall allow specific user to active expired instrument.		Configuration	
complete			4.8.1.6.3		MARK	Henkel need to check NCMR number is not require for instrument Henkel: NCMR is not required.			
complete			4.8.1.7			The system shall have capability to maintain master data for consumable, spare parts, equipment, etc.	Must		
complete	Stock		4.8.1.7.1	Phase A		System shall have capability to maintain master data for consumable, spare parts, equipment.		Standard	
complete			4.8.2.1		Personnel	Configure roles so that Administrator has control of user accounts / roles etc. Lab super user(s) control lab static data…	Must		
complete	Security		4.8.2.1.1	Phase A		System shall allow specific user to maintain user accounts, roles and groups.		Standard	
complete	Security		4.8.2.1.2	Phase A		System shall allow specific user to maintain lab static data.		Standard	
complete			4.8.2.2			The system shall be configurable to allow the group of users to access to data only under his groups.	Must		
complete	Security		4.8.2.2.1	Phase A		System shall allow the group of users to access to data only under his groups.		Standard	
complete	Security		4.8.2.2.2	Phase A		System shall support to configure user's group.		Standard	
complete			4.8.2.3			The system shall provide a way to manage QC technician testing skill against analysis, instrument and preparation including, testing result and testing skill expiry date. This should be based Henkel' s Skill matrix module	Must		
			4.8.2.3.1			System shall support to manage user's training skill by analysis or instrument.		Out Of Scope	
complete	Interface		4.8.2.3.2	Phase B		System shall sync user' s testing skill from Henkel skill matrix tool.		Interface	
complete			4.8.2.3.3		MARK	skill expiry date out of scope in qclad? Henkel: Out of scope		Out Of Scope	
complete			4.8.3.1		SOP	The system shall build a SOP library including description, method version, method ID and other fields. SOP' s shall have version control.	Must		
complete	LES		4.8.3.1.1	Phase B		System shall support to maintain LES method, including description, method version, method ID.		Standard	

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Document Ref: Appendix 01_Functional Requirements_LIMS - Thermo Fisher
Version: 1
Date: 04 June 2024

gathering status	Module	Req. No.	URS Ref.	Delivery Phase	Functional Area / Workshop	Requirement	Priority	Type	Estimate
complete	LES		4.8.3.1.2	Phase B		System shall support version control for LES methods.		Standard	
complete			4.8.3.1.3		Mark	Henkel need to list other fields. Henkel: TF decide the fields required.			
complete			4.8.3.2			The system shall have the ability to hold and generate custom calculations	Must		
complete	LES		4.8.3.2.1	Phase B		System shall be able to maintain custom calculations.		Configuration	
complete			4.8.3.3			SOP should have an approval process and only one version should be active	Must		
complete	LES		4.8.3.3.1	Phase B		System shall support to upgrade version after approval.		Standard	
complete	LES		4.8.3.3.2	Phase B		System shall have only one version active.		Standard	
complete			4.8.4.1		Specification management	Support for multi-level Product Specifications, which must align with product SAP setting	Must		
complete	Sample		4.8.4.1.1	Phase A		System shall support for multi-level Product Specifications.		Standard	
complete	Interface		4.8.4.1.2	Phase A		System shall align MLP data with SAP.		Interface	
complete			4.8.4.2			For non-standard testing, Version controlled Product Specifications that require approval before activation.	Must		
complete	Sample		4.8.4.2.1	Phase A		System shall support to add approval plan for non-standard testing MLP, which requires approval before activation.		Configuration	
complete			4.8.5.1		Stock & Inventory management	Includes information about its characteristics (labels, location, available amount, etc), ordering (supplier, expiry warning, minimum quantity (safety stock) alert, etc), inventory, parameters for use in calculations, etc. Stock in QCLaD, including retain samples, consumables, standards, reagents,	Must		
complete	Stock		4.8.5.1.1	Phase A		System shall support stock mangement to maintain informations including characteristics (labels, location, available amount), ordering (supplier, expiry warning, minimum quantity (safety stock) alert), inventory, parameters for use in calculations.		Configuration	
complete	Stock		4.8.5.1.2	Phase A		System shall support stock mangement to maintain stock type including retain samples, consumables, standards, reagents, instruments, spare parts.		Configuration	
complete			4.8.5.1.3		Mark	what parameters do we need ro use in calculations? Need to check with Henkel team. HENKEL:The same as 4.8.5.4.			
complete	Stock		4.8.5.1.4	Phase A		System shall be able to set and configurate minimum quantity in stock mangement.		Standard	
complete	Stock		4.8.5.1.5	Phase A		System shall be able to set and configurate maximum quantity in stock mangement.		Customisation	
complete	Stock		4.8.5.1.6	Phase A		System shall support to print stock label.		Standard	
complete	Stock		4.8.5.1.7	Phase A		System shall automatically deduct the stock quantity when consumption.		Configuration	
complete			4.8.5.2			The system shall provide a way to track the reference materials used (reference materials and stock solutions) to perform the test(s) on	Must		
complete	LES		4.8.5.2.1	Phase B		System shall allow user to record the reference materials used (reference materials and stock solutions) when performing the tests through barcode.		Configuration	

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Document Ref: Appendix 01_Functional Requirements_LIMS - Thermo Fisher
Version: 1
Date: 04 June 2024

gathering status	Module	Req. No.	URS Ref.	Delivery Phase	Functional Area / Workshop	Requirement	Priority	Type	Estimate
complete			4.8.5.3			The system shall provide a way of tracking the storage location of samples stored in QC by providing material, batch information.	Must		
complete	Retain		4.8.5.3.1	Phase A		System shall allow user to record and retrieve the storage location of samples stored in QC by providing material, batch information.		Configuration	
complete			4.8.5.4			The system shall provide a way for users to record the preparation of stock solutions from one (1) or more than one (1) consumable materials within the inventory. These prepared stock solutions will also be uniquely identified by the system and given an expiry date.	Must		
complete	LES		4.8.5.4.1	Phase B		System shall allow users to prepare stock solutions from one or more consumable materials within the inventory in LES and automatically deduct the consumption and generate a label.		Configuration	
complete	LES		4.8.5.4.2	Phase B		System shall be able to maintain an expiry date and uniquely identify for prepared stock solutions.		Configuration	
			4.8.5.4.3		Mark	What's meaning of the stock solutions? Henkel: System shall automatically deduct the consumption and generate a label clarify expiration date for the new reagent and also be shown in stock inventory.			
complete			4.8.5.5			The system shall provide a way to maintain and view the expiry date record of each consumable material	Must		
complete	Stock		4.8.5.5.1	Phase A		System shall be able to maintain the expiry date field of stock.		Standard	
complete	Stock		4.8.5.5.2	Phase A		System shall be able to display the expiry date field of stock in folder.		Configuration	
complete			4.8.5.6			Warning for expiry and near to the minimum storage level	Must		
complete	Stock		4.8.5.6.1	Phase A		System shall be able to warining for expiry stock in folder.		Standard	
complete	Stock		4.8.5.6.2	Phase A		System shall be able to warining for Low amount storage in folder.		Standard	
complete	Stock		4.8.5.6.3	Phase A		System shall be able to warining for Exceed amount storage in folder.		Customisation	
complete			4.8.5.6.4		MARK	Noted: dragon need maximum amount reminder HENKEL:Reminder for minimum and maximum stock based on the master data. TF added in rtm			
complete			4.8.5.7			The system shall provide a way to automatically assign the validation of chemical reference standards/instrument maintenance task to specified user(s).	Must		
complete			4.8.5.7.1		MARK	Is instrument maintenance out of scope ? Henkel:The same as 4.8.1.2.This one should be in scope.			
complete	LES		4.8.5.7.2	Phase B		System shall create LES tasks for solution preparations automatically when stock is below safety stock or about to expire.		Customisation	
complete	LES		4.8.5.7.3	Phase B		System shall create LES tasks for instrument maintenance automatically when maintenance date arrives.		Configuration	
complete	LES		4.8.5.7.4	Phase B		System shall automatically assign the solution preparation LES tasks to specific user.		Customisation	
complete	LES		4.8.5.7.5	Phase B		System shall automatically assign the instrument maintenance LES tasks to specific user.		Configuration	

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Document Ref: Appendix 01_Functional Requirements_LIMS - Thermo Fisher
Version: 1
Date: 04 June 2024

gathering status	Module	Req. No.	URS Ref.	Delivery Phase	Functional Area / Workshop	Requirement	Priority	Type	Estimate
complete			4.8.5.7.6		MARK	automatically assign rule, validation workflow			
complete			4.8.5.8			Includes handling and storage precautions, security data, associated hazards and risks and GHS requirement.	Must		
complete	Stock		4.8.5.8.1	Phase A		System shall support to maintain stock information , including handling and storage precautions, security data, associated hazards and risks and GHS requirement.		Configuration	
complete	Interface		4.8.5.8.2	Phase A		System shall sync GHS information form SAP to substance in QCLAD. Refer to 4.5.1.5		Out Of Scope	
complete			4.9.1			Security, including audit trails, group and functional security management facilities, password protection of all login accounts. For Electronic Signature support, password checking is required when creating, modifying or deleting electronic records.	Must		
complete	Security		4.9.1.1	Phase A		System shall support function for security, including audit trails, group and security management, password protection of all login accounts.		Standard	
complete			4.9.1.2		MARK	System shall support electronic signature, password checking is required when creating, modifying or removing electronic records.（系统支持在创建、修改、移除记录时触发电子签名并输入密码）		Out Of Scope	
complete			4.9.2			The system shall provide multi layered security allowing different user profiles and access privileges. The related access rights to be configured according to the role performed (e.g. system administrator, system user). Individual users may be assigned one or more profiles or individual access rights.	Must		
complete	Security		4.9.2.1	Phase A		System shall support to configure multiple roles for user to manage permission to profiles and access privileges.(一个人可以有多个角色）		Standard	
complete	Security		4.9.2.2	Phase A		System shall support to maintain different roles.		Standard	
complete			4.9.3			The system shall provide tools to configure access controls at a dataset level or data group level, to restrict viewing and retrieval access to certain individuals or groups	Must		
complete	Security		4.9.3.1	Phase A		System shall support configure access controls using role and master menu.(系统通过角色和菜单按钮控制可见权限）		Standard	
complete			4.9.4			The system must provide unique ID and password authentication processes to confirm and provide traceability to the identity of users. The password must be encrypted or obscured when displayed on screen. This is applicable only for non SSO users	Must		
complete	Security		4.9.4.1	Phase A		System shall support maintain user account id and password. User id shall be unique.（一个人必须有一个唯一的账号，且有密码）		Standard	
complete	Security		4.9.4.2	Phase A		System shall encrypt or obscured password when displayed on screen.（密码是加密的）		Standard	
complete			4.9.5			Single Sign On has to be integrated and used for all users with Henkel user ids. For Henkel operators, they will be accessing application through iPAD with application credentials and password as defined in the above.	Must		
complete	Security		4.9.5.1	Phase A		System shall support single sign on for all users.		Standard	

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Document Ref: Appendix 01_Functional Requirements_LIMS - Thermo Fisher
Version: 1
Date: 04 June 2024

gathering status	Module	Req. No.	URS Ref.	Delivery Phase	Functional Area / Workshop	Requirement	Priority	Type	Estimate
complete			4.9.5.2		Mark	Henkel user id是想用windows汉高的密码直接登录QCLAD吗? Henkel: Only for white collar is possible.For blue collar, they need user ID and password for QCLaD			
complete	Security		4.9.5.3	Phase A		System shall allow user to login through iPad with user id and password.		Standard	
complete	Security		4.9.5.4	Phase A		System shall be able to limit user to login QCLAD using one account for one device or cilent at one time.		Standard	
complete			4.9.5.5		MARK	HENKEL: one account login to system in multiple devices at same time may required . Need further discuss internally （待确认一个账户是否要能同时登录多设备）		Standard	
complete			4.9.6			User accounts must be suspended after a configurable number of consecutive unsuccessful attempts to enter a password into the QCLAD system until a Security Administrator resets the password	Must		
complete	Security		4.9.6.1	Phase A		System shall be able to freeze user's account after configurable login failure attempts. （多次输错密码冻结账户）		Standard	
complete	Security		4.9.6.2	Phase A		System shall allow specific user to unlock the account or reset the password.（管理员能够解锁和重置密码）		Standard	
complete			4.9.7			The system shall provide Administrator level account holders with the ability to terminate a user session on the system	Must		
complete	Security		4.9.7.1	Phase A		System shall support system Administrator to terminate a user session on the system.(管理员终止用户进程）		Configuration	
complete			4.9.8			The system shall provide global admin/user, regional admin/user and site admin/user roles. Each plant should work independently, and their activities, capabilities based on the plant level.	Must		
complete	Security		4.9.8.1	Phase A		System shall support maintain global admin/user role.		Standard	
complete	Security		4.9.8.2	Phase A		System shall support maintain regional admin/user role.		Standard	
complete	Security		4.9.8.3	Phase A		System shall support maintain site admin/user role.		Standard	
complete	Security		4.9.8.4	Phase A		System shall support to work independently, perform activities, capabilities at plant level.		Standard	
complete			4.9.8.5		MARK	1、TF Give an example of Site permissions division 2、Henkel Provide a list of permissions for a role, such as the permissions and scope of data that the role needs to view/modify; 1、TF 给个Site权限划分的example 2、Henkel 提供角色的权限清单，例如该角色需要查看/修改数据的权限和范围;			

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Document Ref: Appendix 01_Functional Requirements_LIMS - Thermo Fisher
Version: 1
Date: 04 June 2024

gathering status	Module	Req. No.	URS Ref.	Delivery Phase	Functional Area / Workshop	Requirement	Priority	Type	Estimate
			4.10.1		Project management	•Responsible for project management for all in scope work including qualified project team set up, project schedule, timeline, milestone, deliverables, implementation, cooperation with HENKEL and relevant contractors, etc. •Roles and responsibilities of different parties •Expectations from Henkel has to be defined, documented after agreement between both parties •Requirements to be signed off between both parties •Escalation rule should be clearly defined for global scope. •All communication to plant team strictly with the core team members knowledge. Any other communication without involving core team leads, will not be considered legitimate. •Dragon pilot and global roll out should be managed by same global team with their regional/country counterparts.	Must	Business Proc Only	
			4.10.1.1						
			4.10.2		Implementation	•Cooperate with HENKEL to analyse, implement and customize the system for HENKEL •Software installation and hardware landscape establishment •Implementation and delivery of a system that fully integrated with Henkel Systems •Assisting HENKEL to establish solution and support technical issue & final configuration, also assisting to establish testing scenarios and user acceptance test •Remedy test problems •Implementation of necessary program changing to fit for purpose work processes and Henkel QCLAD template •System vendor should recommend required hardware type and brand to Henkel based on Henkel local site condition	Must	Business Proc Only	
			4.10.2.1						
			4.10.3.1		Training	•Initial training to key users for pre-project preparations with a set of training materials	Must	Business Proc Only	
			4.10.3.1.1						
			4.10.3.2			•Training language should be available in English or user preferred local language based on the language preference of the plant to be rolled out.	Must	Business Proc Only	
			4.10.3.2.1						
			4.10.4		License	•Providing all necessary software licenses •Software license purchased by each party has to be well defined	Must	Business Proc Only	
			4.10.4.1						
			4.10.5		Post support/Hyper care	•Providing support which detail SLA specified in MA contract •Providing post go-live maintenance and support •Roles and responsibilities of different parties during post Go live, roll out and maintenance. •Hypercare should be conducted after system go live, system vendor should be support 60 man days to make the system running smoothly at Henkel lobal sites.	Must	Business Proc Only	
			4.10.5.1						

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Document Ref: Appendix 01_Functional Requirements_LIMS - Thermo Fisher
Version: 1
Date: 04 June 2024

gathering status	Module	Req. No.	URS Ref.	Delivery Phase	Functional Area / Workshop	Requirement	Priority	Type	Estimate
			4.10.6		Project documents	•Providing all relevant documentations with document categorise detailed •Provide system user manual in English and Henkel site local language	Must	Business Proc Only	
			4.10.6.1						
			4.11.1						
	Dashboard		4.11.1.1			需要有两个仪表盘，一个为带有过滤条件的静态仪表盘，一个是默认条件但是定时刷新的动态仪表盘：两个仪表盘中显示的内容一致		Customisation	
	Dashboard		4.11.1.2			带有过滤条件的静态仪表盘的过滤条件为：plant, work center, business team(group)		Customisation	
	Dashboard		4.11.1.3			默认条件的动态仪表盘的过滤条件为：当前登录人员所在的plant, work center, business team(group)		Customisation	
	Dashboard		4.11.1.4			仪表盘中需显示：Date of the day、of that day, the lists of on-duty operators, only showing the operator ID		Customisation	
	Dashboard		4.11.1.5			仪表盘中需显示：of the criteria, the statistic of all the tasks ： 当天收样但未审核完成的样品数量 当天之前收样但未审核完成的样品数量 测试已完成的样品数量completed 测试中的样品数量in-testing 未开始的样品数量not started the count of delayed sample the count of urgent sample		Customisation	
	Dashboard		4.11.1.6			仪表盘中需显示：the list of delayed sample		Customisation	
	Dashboard		4.11.1.7			仪表盘中需显示：the list of urgent sample		Customisation	
	Dashboard		4.11.1.8			仪表盘中需显示：the column chart in which x axis has the Sample Count, and y axis has the product line.且柱状图中按照样品状态（未开始，测试中，测试已完成）分颜色显示		Customisation	
	Dashboard		4.11.1.9			仪表盘中需显示：anotther column chart in which x axis has the Sample Count, and y axis has the inspection type (IQC, OQC, IPQC)，且柱状图中按照样品状态（未开始，测试中，测试已完成）分颜色显示		Customisation	

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Document Ref: Appendix 01_Functional Requirements_LIMS - Thermo Fisher
Version: 1
Date: 04 June 2024

gathering status	Module	Req. No.	URS Ref.	Delivery Phase	Functional Area / Workshop	Requirement	Priority	Type	Estimate

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Document Ref: Appendix 01_Functional Requirements_LIMS - Thermo Fisher
Version: 1
Date: 04 June 2024

gathering status	Module	Req. No.	URS Ref.	Delivery Phase	Functional Area / Workshop	Requirement	Priority	Type	Estimate

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Document Ref: Appendix 01_Functional Requirements_LIMS - Thermo Fisher
Version: 1
Date: 04 June 2024

gathering status	Module	Req. No.	URS Ref.	Delivery Phase	Functional Area / Workshop	Requirement	Priority	Type	Estimate

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Document Ref: Appendix 01_Functional Requirements_LIMS - Thermo Fisher
Version: 1
Date: 04 June 2024

gathering status	Module	Req. No.	URS Ref.	Delivery Phase	Functional Area / Workshop	Requirement	Priority	Type	Estimate

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Document Ref: Appendix 01_Functional Requirements_LIMS - Thermo Fisher
Version: 1
Date: 04 June 2024

gathering status	Module	Req. No.	URS Ref.	Delivery Phase	Functional Area / Workshop	Requirement	Priority	Type	Estimate

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